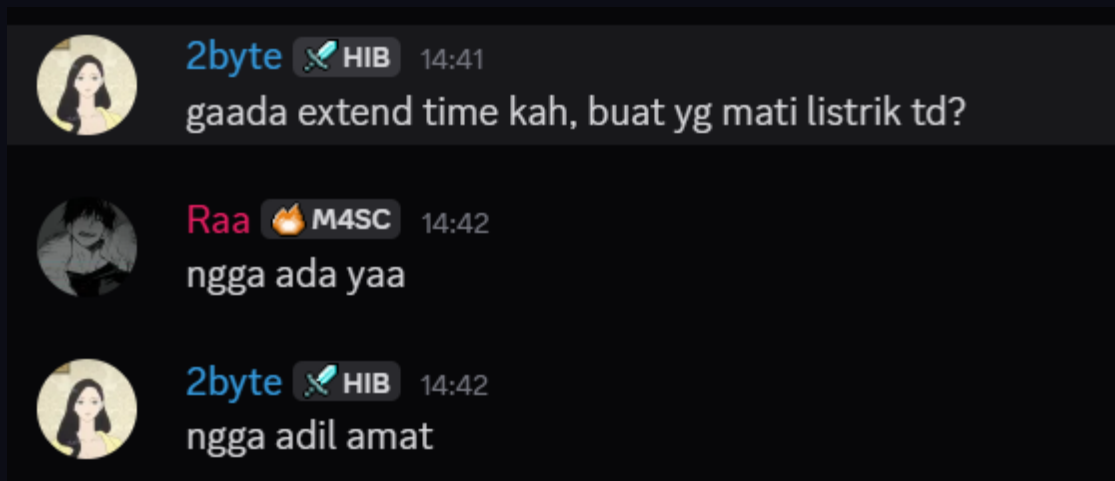




Write Up Finals

GCW 2.0

Team **balik nama HiB - ANT TECH**
ANT TECH ASYNC lah, ga hoki njir
pake nama poros akwoakwoakwoak



2byte . AriqArdian . zenc

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Cryptography

Recovery (100 Points)

three traces remain... the secret is waiting to be recovered.

Author: wielio

Jadi pada challenge ini kita diberikan sebuah file yaitu [chall.py](#)

```

1  import os
2  import random
3  import hashlib
4  from Crypto.Cipher import AES
5  from secret import FLAG, M, K      Import "secret" could not be resolved
6  p = 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFC2F
7
8  def srng(state):
9      return (state * M + K) % p
10
11  secret = random.randint(2, p - 2)
12  seed = random.randint(2, p - 2)
13  session_id = srng(seed)
14  a1 = srng(session_id)
15  a2 = srng(a1)
16  trace_alpha = srng(a2)
17  trace_bravo = srng(trace_alpha)
18  trace_charlie = srng(trace_bravo)
19
20  def f(x):
21      return (secret + a1 * x + a2 * x * x) % p
22
23  # Node shares
24  y_bravo = f(2)
25  y_charlie = f(3)
26  flag_key = hashlib.sha256(secret.to_bytes(32, 'big')).digest()
27  flag_nonce = os.urandom(16)
28  cipher = AES.new(flag_key, AES.MODE_GCM, nonce=flag_nonce)
29  flag_ct, flag_tag = cipher.encrypt_and_digest(FLAG)

```

Dan disini tipe nya ada random parameter file , dan disini juga diberikan beberapa parameter output dari soal yaitu.

```

1 session_id = 7160730235885832832919867569460157812911626607030223710101453324690485374888
2 trace_alpha = 61440536045064128691097134029847638761687802416384796192237589001009960971122
3 trace_bravo = 37315296588380358445482742325820468750320938390564468768025417677942521718991
4 trace_charlie = 20617629846758726153982233326380232750917323510355011808011337963071941558209
5 share_bravo = 33256473601104795798385995777409145093413187116532670737846058558004098140891
6 share_charlie = 112543661875884619325516411388117612905773101713566944502814775978065019461248
7 flag_nonce = '84dad38d62de66ab34952ca6177b68db'
8 flag_tag = '5125d12dd6ba9fc8227e9e023bc659e7'
9 flag_ct = 'ebb5931113b3beeb148a34cf6164f9669200eeddee5552e5e347de1c63fa744b560cfd6bd9930a07cebaefc2d2410aff3a'
10

```

Lalu disini untuk mencari nilai dari parameter seperti a-1 dan a-2 adalah

- $a_1 = \text{srng}(\text{session_id}) = (\text{session_id} \cdot M + K) \bmod p$
- $a_2 = \text{srng}(a_1) = (a_1 \cdot M + K) \bmod p$
- $\text{trace_alpha} = \text{srng}(a_2) = (a_2 \cdot M + K) \bmod p$
- $\text{trace_bravo} = \text{srng}(\text{trace_alpha}) = (\text{trace_alpha} \cdot M + K) \bmod p$

Lalu untuk mencari nilai dari M & K adalah

```

numerator = (trace_bravo - trace_alpha) % p
denominator = (trace_alpha - trace_bravo) % p

denom_inv = modular_inverse((trace_alpha - trace_bravo) % p, p)
M = ((trace_bravo - trace_charlie) * denom_inv) % p
K = (trace_bravo - (trace_alpha * M)) % p

```

Lalu disini untuk mencari nilai A nya adalah menggunakan function dari srng lalu menghitung session_id dan untuk a2 menggunakan nilai a1.

```

def srng(state):
    return (state * M + K) % p

a1 = srng(session_id)
a2 = srng(a1)

print(f"[+] Computed a1: {a1}")
print(f"[+] Computed a2: {a2}")

```

Lalu rumus untuk mencari nilai secretnya adalah.

```
secret = (y_bravo - 2*a1 - 4*a2) % p

print(f"[+] Recovered Secret: {secret}")
```

jadi $\text{secret} = (y_{\text{bravo}} - 2(a_1) - 4(a_2) \bmod p$

setelah di dapatkan nilai secret, nilai secret di convert ke sha 256 untuk di jadikan key AES

```
flag_key = hashlib.sha256(secret.to_bytes(32, 'big')).digest()

try:
    cipher = AES.new(flag_key, AES.MODE_GCM, nonce=flag_nonce)
    decrypted_flag = cipher.decrypt_and_verify(flag_ct, flag_tag)
    print(f"[+] Decrypted Flag: {decrypted_flag.decode('utf-8')}")
except ValueError as e:
    print(f"\n[!] Decryption failed: {e}")
```

Dan setelah itu hasil dari key nya digunakan untuk decrypt dari algoritma AES GCM dan ketika berhasil dijalankan akan mengeluarkan flag

```
AriqArdian : ~/C/GC/Finals/Cry/Recovery → python solve2.py
[+] Found M: 97791389573415822071767189154718148724150916242148418035132190969352491429918
[+] Found K: 57785545891542916597005168815843022135840852446702717521906999692697291038385
[+] Computed a1: 31969543785495612630051985741143492849951292460418729402528705372320928318295
[+] Computed a2: 32621946745320081264129882975650576563135721360451221680379519211129765534745
[+] Recovered Secret: 70413777523465636328904462409895668847507686085171453290185738984660848708647
[+] Decrypted Flag: GCW{d373rm1n1571c_5rn6_l34k_v14_7r4c3_1d5_5h4m1r}
```

FLAG: GCW{d373rm1n1571c_5rn6_l34k_v14_7r4c3_1d5_5h4m1r}

Digital Forensic

Silent Execution (50 Points)

```
ss soal
```

Dikasih memdump windows dan langsung aja pake plugin windows.cmdline di volatility karena ini terkait silent execution yang biasanya didapat dari cmd line.

```
9984 powershell.exe "C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe" -NoProfile -WindowStyle Hidden -EncodedCommand JABmACAAPQAgACIARwBDAFcAewBwADAAdwAzAHIAcwBoADMABABsAF8AMwBuAGMAMABkADMAZABfAGIANABzADMANgA0AF8AMQBwAF8AUgBBAE0AFQAIADsAIABXAHIAaQB0AGUALQBPAHUAdABwAHUAdAAgACQAZgA7ACAAUwB0AGEAcgB0AC0AUwBsAGUAZQBwACAALQBTAGUAYwBvAG4AZABzACAAOQA5ADkAOQA5AA==
4448 conhost.exe \??\C:\WINDOWS\system32\conhost.exe 0x4
6996 svchost.exe C:\WINDOWS\system32\svchost.exe -k GPSvcGroup
7904 go-winpmem_amd "C:\Users\victi\Downloads\go-winpmem_amd64_1.0-rc2_signed.exe" acquire chall.raw
```

Di akhir-akhir, ada base64 panjang dan sesuai deskripsi hal ini dijalankan secara hidden (silent).

```
(byte@byuda)-[mnt/c/Users/ACER/Downloads/CTF/gcwug/foren/silent]
$ echo "JABmACAAPQAgACIARwBDAFcAewBwADAAdwAzAHIAcwBoADMABABsAF8AMwBuAGMAMABkADMAZABfAGIANABzADMANgA0AF8AMQBwAF8AUgBBAE0AFQAIADsAIABXAHIAaQB0AGUALQBPAHUAdABwAHUAdAAgACQAZgA7ACAAUwB0AGEAcgB0AC0AUwBsAGUAZQBwACAALQBTAGUAYwBvAG4AZABzACAAOQA5ADkAOQA5AA==" | base64 -d
$f = "GCW{p0w3rsh3ll_3nc0d3d_b4s364_1n_RAM}"; Write-Output $f; Start-Sleep -Seconds 9999
```

FLAG: GCW{p0w3rsh3ll_3nc0d3d_b4s364_1n_RAM}

Reze Was Here (468 Points)

ss soal

Dikasih image windows langsung aja load ke autopsy. Q1 itu ditanyaain user nya siapa, jawabannya simple ada di C:\Users\(nama)

Listing						
/img_SCADA-WS04.img/vol_vol3/Users						
Table Thumbnail Summary						
Name	S	C	O	Modified Time	Change Time	
 analyst				2026-06-18 17:35:08 WIB	2026-06-18 17:35:08 WIB	
 All Users				2019-12-07 16:30:39 WIB	2026-06-19 08:16:08 WIB	
 Default User				2019-12-07 16:30:39 WIB	2026-06-19 08:16:08 WIB	
 desktop.ini			0	2019-12-07 16:12:42 WIB	2026-06-19 08:15:08 WIB	
 Public				2026-06-19 01:12:31 WIB	2026-06-19 01:12:31 WIB	
 [current folder]				2026-06-18 17:48:46 WIB	2026-06-18 17:48:46 WIB	

Q2 hostname nya adalah scada-ws04 dapet dari os info di autopsy

Listing						
Operating System Information						
Table Thumbnail Summary						
Source Name	S	C	O	Name	Program Name	Processor Architecture
 SCADA-WS04.img				SCADA-WS04	Windows 10 Pro	AMD64

Q3 didapet dari dir C:\Users\Public\ dan ada sebanyak 10 .blf file

Listing						
/img_SCADA-WS04.img/vol_vol3/Users/Public						
Table	Thumbnail	Summary				
Name	S	C	O	Modified Time	Change Time	
 2825_9.blf			0	2026-06-19 01:12:20 WIB	2026-06-19 0	
 p_nk			0	2026-06-19 01:12:20 WIB	2026-06-19 0	
 2825_6.blf			0	2026-06-19 01:12:19 WIB	2026-06-19 0	
 2825_7.blf			0	2026-06-19 01:12:19 WIB	2026-06-19 0	
 2825_8.blf			0	2026-06-19 01:12:19 WIB	2026-06-19 0	
 2825_3.blf			0	2026-06-19 01:12:18 WIB	2026-06-19 0	
 2825_4.blf			0	2026-06-19 01:12:18 WIB	2026-06-19 0	
 2825_5.blf			0	2026-06-19 01:12:18 WIB	2026-06-19 0	
 2825_1.blf			0	2026-06-19 01:12:17 WIB	2026-06-19 0	

Q4 didapat dari sana juga dan namanya adalah 2825.blf

Listing						
/img_SCADA-WS04.img/vol_vol3/Users/Public						
Table	Thumbnail	Summary				
Name	S	C	O	Modified Time	Change Time	
 2825.blf			0	2026-06-19 01:12:16 WIB	2026-06-19 0	
 AccountPictures				2026-06-18 17:32:14 WIB	2026-06-18 1	
 desktop.ini			0	2019-12-07 16:12:42 WIB	2026-06-19 0	
 [current folder]				2026-06-19 01:12:31 WIB	2026-06-19 0	
 Desktop				2026-06-19 07:17:34 WIB	2026-06-18 1	
 Documents				2026-06-19 07:18:32 WIB	2026-06-19 0	
 Downloads				2019-12-07 16:14:54 WIB	2026-06-19 0	
 Libraries				2019-12-07 16:31:03 WIB	2026-06-19 0	
 Music				2019-12-07 16:14:54 WIB	2026-06-19 0	

Q5 extract dari USN journal, lalu find _svc_

svc		FileCreate	File creations	Renames	Overwrites	Deletions
From	dd / mm / yyyy, --:--	To	dd / mm / yyyy, --:--			
Time	Filename	Full path	REASONS	USN		
2026-06-18 18:12:20.000 UTC	dump_svc_0619	—	FileCreate	14759248		
2026-06-18 18:12:20.000 UTC	dump_svc_0619	—	DataExtend FileCreate	14759336		
2026-06-18 18:12:20.000 UTC	dump_svc_0619	—	DataExtend FileCreate Close	14759424		

Q6 search aja di google pake keyword 2825.blf clfs lalu ketemu cve nya

2825.blf clfs

Core Security
https://www.coresecurity.com > ... > Articles :

Analisis Kerentanan CVE-2023-28252 CLFS

10 Jul 2023 — Kerentanan CVE-2023-28252 menargetkan Common Log File System (CLFS) dan memungkinkan penyerang untuk meningkatkan hak akses dan berpotensi ...

🌐 Diterjemahkan oleh Google [Lihat versi asli \(English\)](#)

Tidak ada: 2825- | Telusuri dengan: 2825.

Q7 umumnya registry credential aja yaitu SAM

Listing										48 Results
/img_SCADA-W504.img/vol_vol3/Windows/System32/config										
Table Thumbnail Summary										
										Save Table as CSV
Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	
ELAM(53b39eac-18c4-11ea-a811-000d3aa4692b).1				2026-06-19 07:17:28 WIB	2026-06-19 07:17:28 WIB	2026-06-19 07:17:28 WIB	2026-06-19 07:17:28 WIB	65536	Allocated	
ELAM(53b39eac-18c4-11ea-a811-000d3aa4692b).1				2026-06-19 07:17:28 WIB	2026-06-19 07:17:28 WIB	2026-06-19 07:17:28 WIB	2026-06-19 07:17:28 WIB	524288	Allocated	
ELAM(53b39eac-18c4-11ea-a811-000d3aa4692b).1				2026-06-19 07:17:28 WIB	2026-06-19 07:17:28 WIB	2026-06-19 07:17:28 WIB	2026-06-19 07:17:28 WIB	524288	Allocated	
SAM				2026-06-19 02:13:27 WIB	2026-06-19 08:16:39 WIB	2026-06-19 02:13:27 WIB	2019-12-07 16:03:44 WIB	65536	Allocated	
SAM.LOG1				2019-12-07 16:03:44 WIB	2026-06-19 08:16:39 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	65536	Allocated	
SAM.LOG2				2019-12-07 16:03:44 WIB	2026-06-19 08:16:42 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	49152	Allocated	
SECURITY				2026-06-19 02:13:27 WIB	2026-06-19 08:16:39 WIB	2026-06-19 02:13:27 WIB	2019-12-07 16:03:44 WIB	32768	Allocated	
SECURITY.LOG1				2019-12-07 16:03:44 WIB	2026-06-19 08:16:39 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	32768	Allocated	
SECURITY.LOG2				2019-12-07 16:03:44 WIB	2026-06-19 08:16:42 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	65536	Allocated	

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

ROOT

SAM

Domains

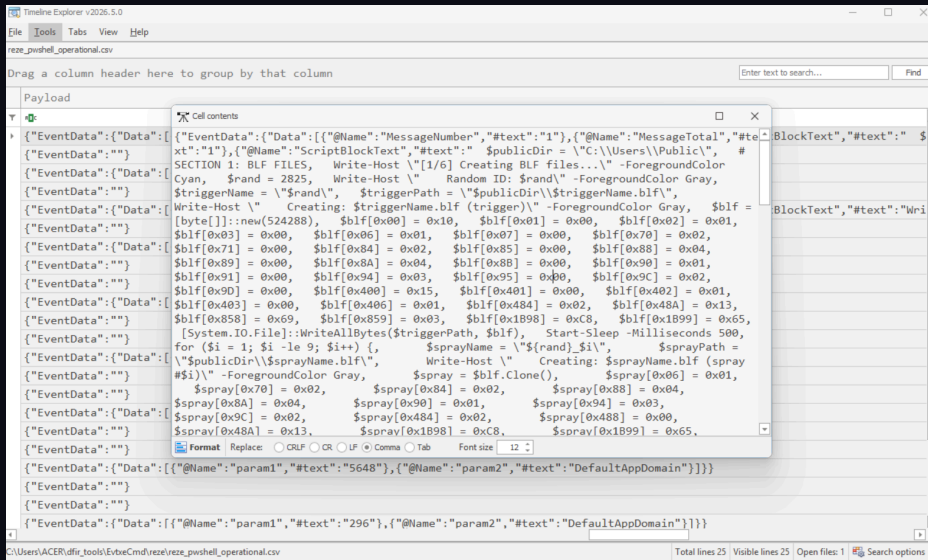
LastSkuUpgrade

RXACT

C

ServerDomainUpdates

Q8 saya dapet referensi di <https://github.com/fortra/CVE-2023-28252> bahwa exploit dijalankan lewat command line. Langsung aja cek logs powershell operational.evtx nya dan parsing pake evtxcmd trus masukin ke timeline explorer. Dapet deh full script exploitnya dan persistence service



```
1 Creating registry persistence...\ -ForegroundColor Cyan, $serviceName = "WSearchMaintenanceSvc", $imageP
```

```
from pwn import *
p = remote('ctf.gcwug.com', 1204)
print(p.recvuntil(b'Answer : ').decode())
answers = [
    'analyst',
    'scada-ws04',
    '10',
    '2825.blf',
    'dump_svc_0619',
    'CVE-2023-28252',
    'SAM',
    'WSearchMaintenanceSvc'
]

for answer in answers:
    p.sendline(answer.encode())
    response = p.recv().decode()
    print(response)
print(p.recvall().decode())
```

FLAG: GCW{r3z3_s4ys_h1_t0_ur_s4m_lol}

// Minimal comment nya hapus mas, probset nge ai tapi peserta ngga boleh ai

Presented by @hackitbraw_ub

Miscellaneous

Lost in Translation (100 Points)

you're smart, you'll figure it out.

Author: zennova

Intinya di challenge ini kita diberikan sebuah program yang memberikan pertanyaan selama 50x dan di pertanyaan itu ada 2 tipe yaitu bisa ubah string ke ascii atau penjumlahan jadi kita disini bikin script otomatisasi saja karena timeout nya cepat dan berikut bentuk scriptnya.

```
from pwn import *
import re

context.log_level = 'info'

r = remote('ctf.gcwug.com', 1644)

try:
    for i in range(50):
        raw_data = r.recvuntil(b'> ')
        line = raw_data.decode()

        print(f"[#] Progress: {i+1}/50")

        if "to hex" in line:
            match_hex = re.search(r"Translate '([^\']+) ' to hex", line)
            if match_hex:
                teks_target = match_hex.group(1)
                jawaban = teks_target.encode('utf-8').hex()
                r.sendline(jawaban.encode())
            else:
                print("[-] Gagal mengekstrak string untuk hex.")
                break

        elif "What is" in line:
            match_math = re.search(r"What is (\d+)\s*\+\s*(\d+)", line)
```

```

        if match_math:
            num1 = int(match_math.group(1))
            num2 = int(match_math.group(2))
            jawaban = num1 + num2
            r.sendline(str(jawaban).encode())
        else:
            print("[-] Gagal mengekstrak angka matematika.")
            break

    else:
        print("[-] Menemukan tipe soal baru yang belum terdaftar!")
        print(f"Isi teks dari server:\n{line}")
        break

    print("\n[+] Berhasil melewati 50 stage! Membaca respons server...")
    print(r.recvall().decode())

except EOFError:
    print("\n[-] Koneksi terputus oleh server (EOFError).")

r.close()

```

Ketika berhasil akan keluar output seperti berikut.

```

[+] Berhasil melewati 50 stage! Membaca respons server...
[+] Receiving all data: Done (158B)
[*] Closed connection to ctf.gcwug.com port 1644
OK

Nice. You survived stage 1.
Stage 2 unlocked: https://drive.google.com/drive/folders/1myUdM9fILGTdgZnnM74oggFHz8Br70bf?usp=sharing
A new language awaits.

```

Ketika link ini di akses terdapat sebuah file [stage2.tar.xz](https://drive.google.com/drive/folders/1myUdM9fILGTdgZnnM74oggFHz8Br70bf?usp=sharing) yang dimana file tersebut, terdapat sebuah teks panjang dan di bawah terdapat sebuah teks berikut.

```

My Esoteric language gw rules aturan

Start with total = 0.

ADD X
    total = (total + X) & 0xff

XOR X
    total = total ^ X

OUT
    print chr(total)

Every line is one instruction.

```

Dan di satu file berikut nya terdapat kode berikut.

```
XOR 70  
OUT  
ADD 38  
OUT  
XOR 13  
OUT  
ADD 6  
OUT  
ADD 185  
OUT  
ADD 82  
OUT  
ADD 243  
OUT  
ADD 252  
OUT  
ADD 11  
OUT  
XOR 76  
OUT  
ADD 66  
OUT  
XOR 7  
OUT  
XOR 11  
OUT  
XOR 11  
OUT  
ADD 13  
OUT  
XOR 19  
OUT  
ADD 13  
OUT  
XOR 78  
OUT  
XOR 83  
OUT  
ADD 242  
OUT  
XOR 23  
OUT  
ADD 247  
OUT  
ADD 6  
OUT  
XOR 26
```

Intinya ini program yang dibuat dari probset nya kalo untuk decrypt nya ikutin aturan dari file yang pertama, Lalu disini aku membuat script nya untuk mendecrypt isi file tersebut.


```

XOR 18
OUT
XOR 58
OUT
XOR 62
OUT
XOR 10
OUT
XOR 30
OUT
ADD 234
OUT
ADD 20
OUT
ADD 238
OUT
ADD 24
OUT
XOR 24
OUT
XOR 28
OUT
"""

print(solve_instructions(data_instruksi))

```

Dan ketika dijalankan menghasilkan output berikut.

```

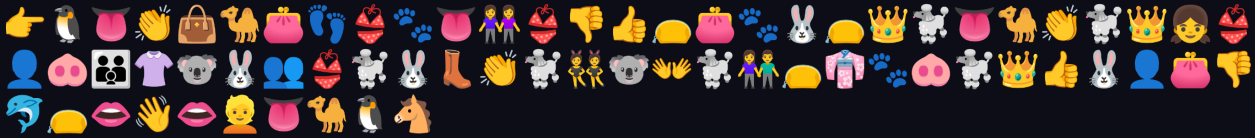
AriqArdian : ~/C/GC/F/C/R/LostTranslation → python solve2.py 6.788s base system
Flag real beneran serious trust me bro oke? oke?? oke gas oke gas mari kita honeymoon ke paris: GCW{when_yh_bisa_ikuta
n_#kaburajadulu_jujur_muak_juga_lama2_gwe_aku_saya}

```

FLAG:GCW{when_yh_bisa_ikutan_#kaburajadulu_jujur_muak_juga_lama2_gwe_aku_saya}

Sanity Check (100 Points)

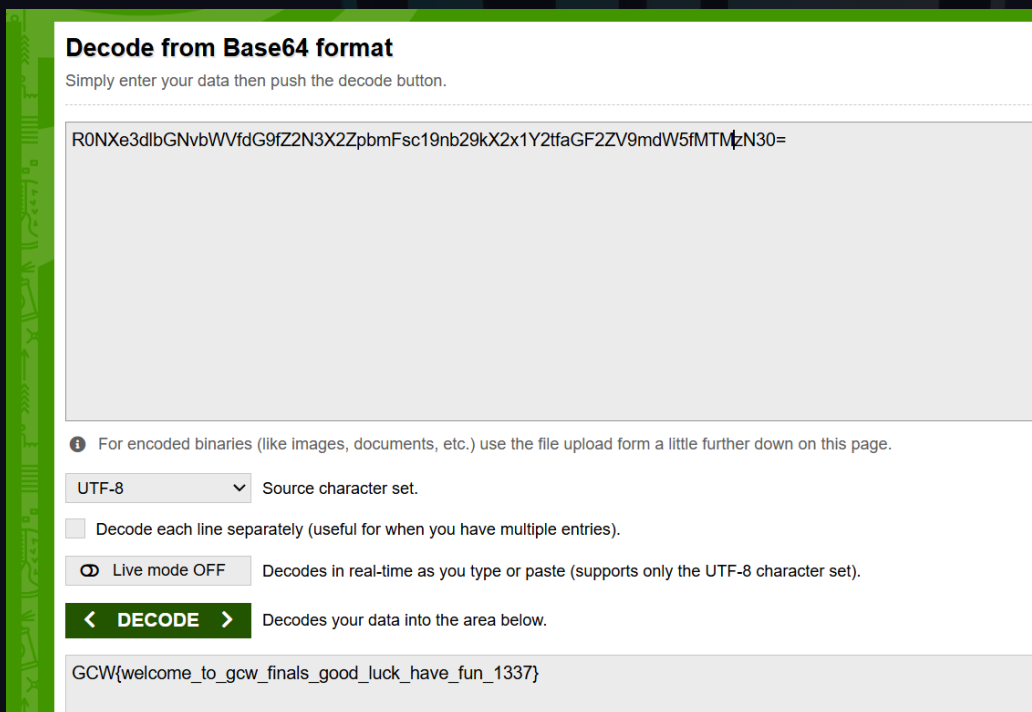
Ibnu Jangan Emote Ibnu



Intinya ini emoji Encrypt jadi yaudah tinggal nyari Emoji encrypt nya di web



Udah nemu decrypt nya dapat hasil nya itu base64 tinggal ke base64 decoder. dan...



Udah Dapet flag nya.

FLAG: GCW{welcome_to_gcw_finals_good_luck_have_fun_1337}

feedback final (15 Points)

Jangan lupa untuk mengisi form feedback yang telah disediakan. Setiap masukan, kritik, dan saran terkait pelaksanaan kompetisi maupun deskripsi soal sangat berarti bagi kami sebagai bahan evaluasi dan pengembangan Gunadarma Code Week ke depannya.

[Feedback](#)

Hujat aja adminnya di form feedback nanti dapat flag. Kecwa berat karena ngga ada prepare yang proper untuk kelistrikan yang jadinya buang - buang waktu.



FLAG: GCW{final_In_J6}